

Ethanol Blending in Gasoline – Greece

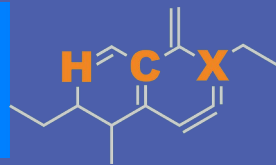
July 2026



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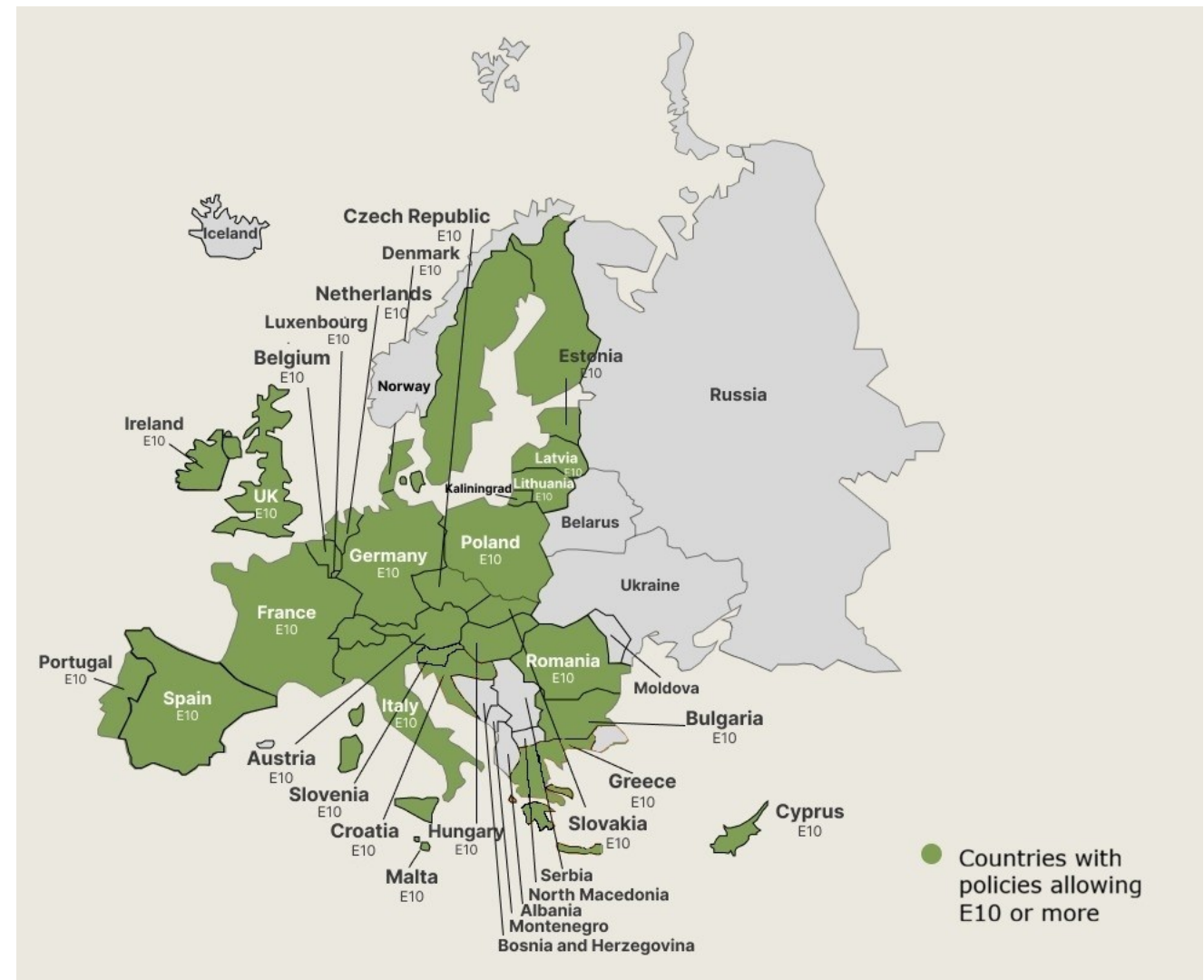


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

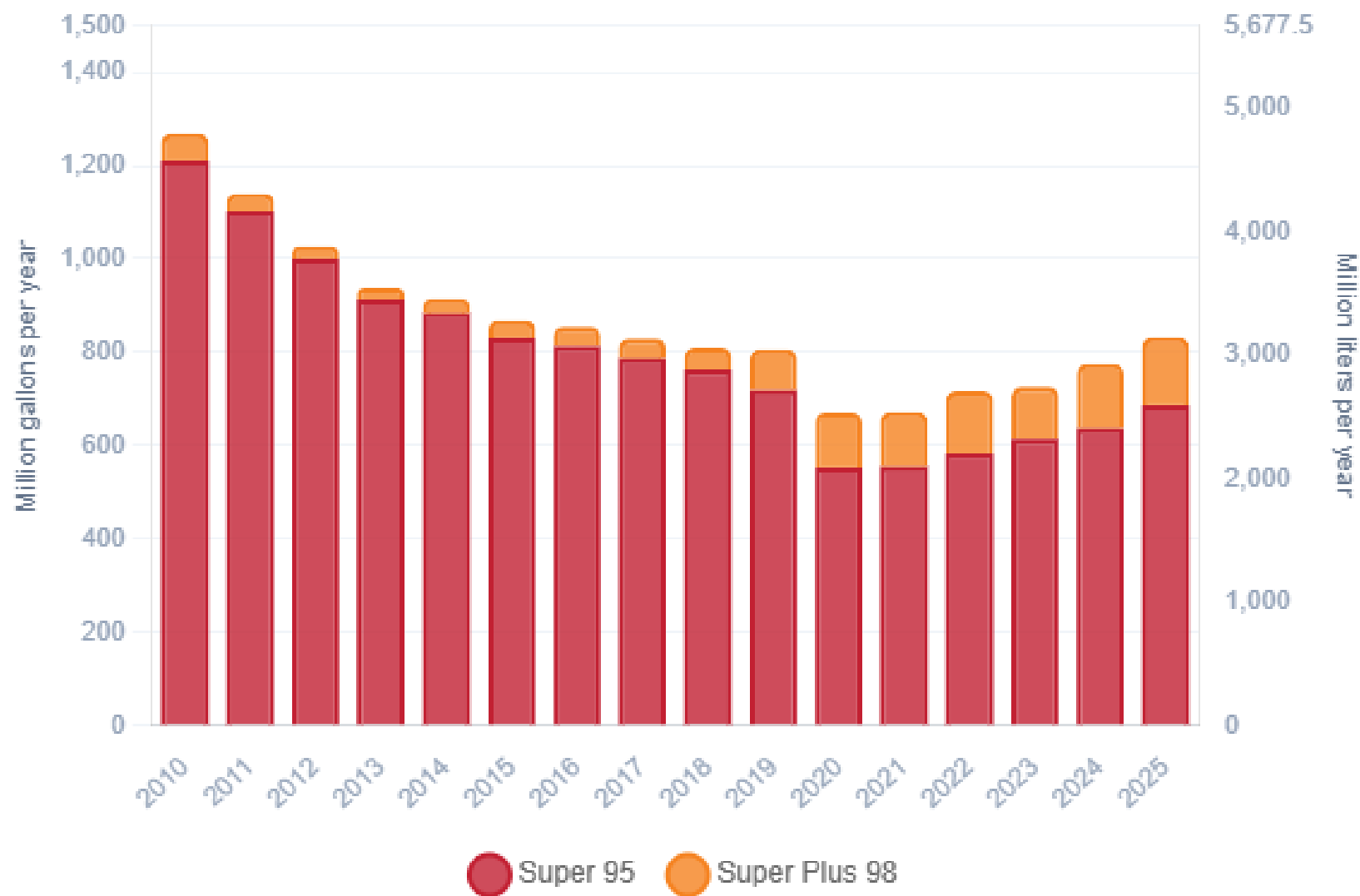
Ethanol Gasoline Blending in Greece



In 2025, gasoline consumption in Greece was 3,156 million liters (834 million gallons) and growth rate was -2.8%. Gasoline production and net imports were 7,983 and -6041 million liters (2,109 and -1,596 million gallons) correspondingly. Greece complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.5 RON.

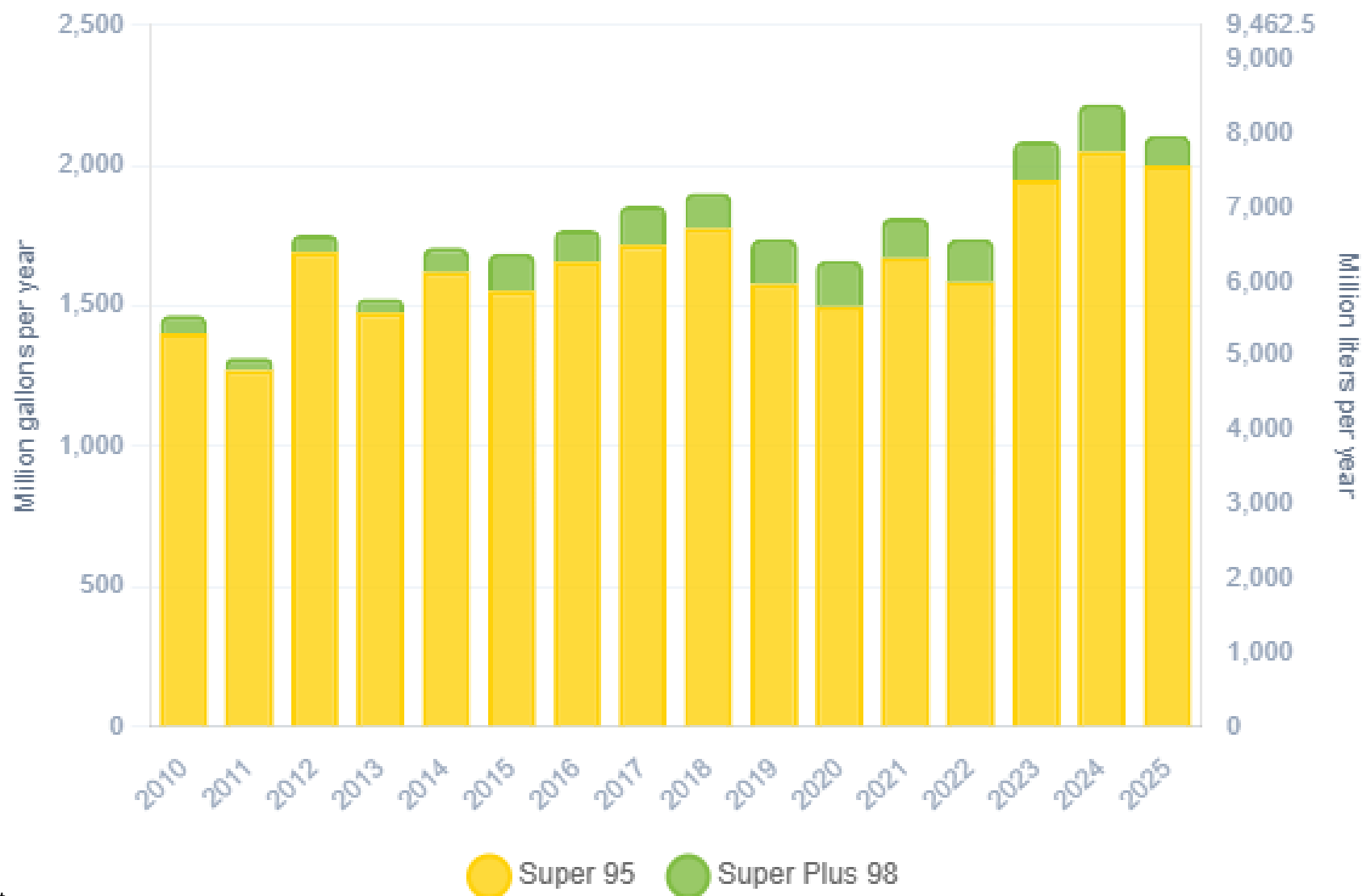
Ethanol consumption in Greece is determined by 3.3% bioethanol content mandate (by energy). In 2025 demand was 164 million liters (43 million gallons) representing 5.0% of gasoline demand. Ethanol is supplied by imports.

Gasoline Demand in Greece



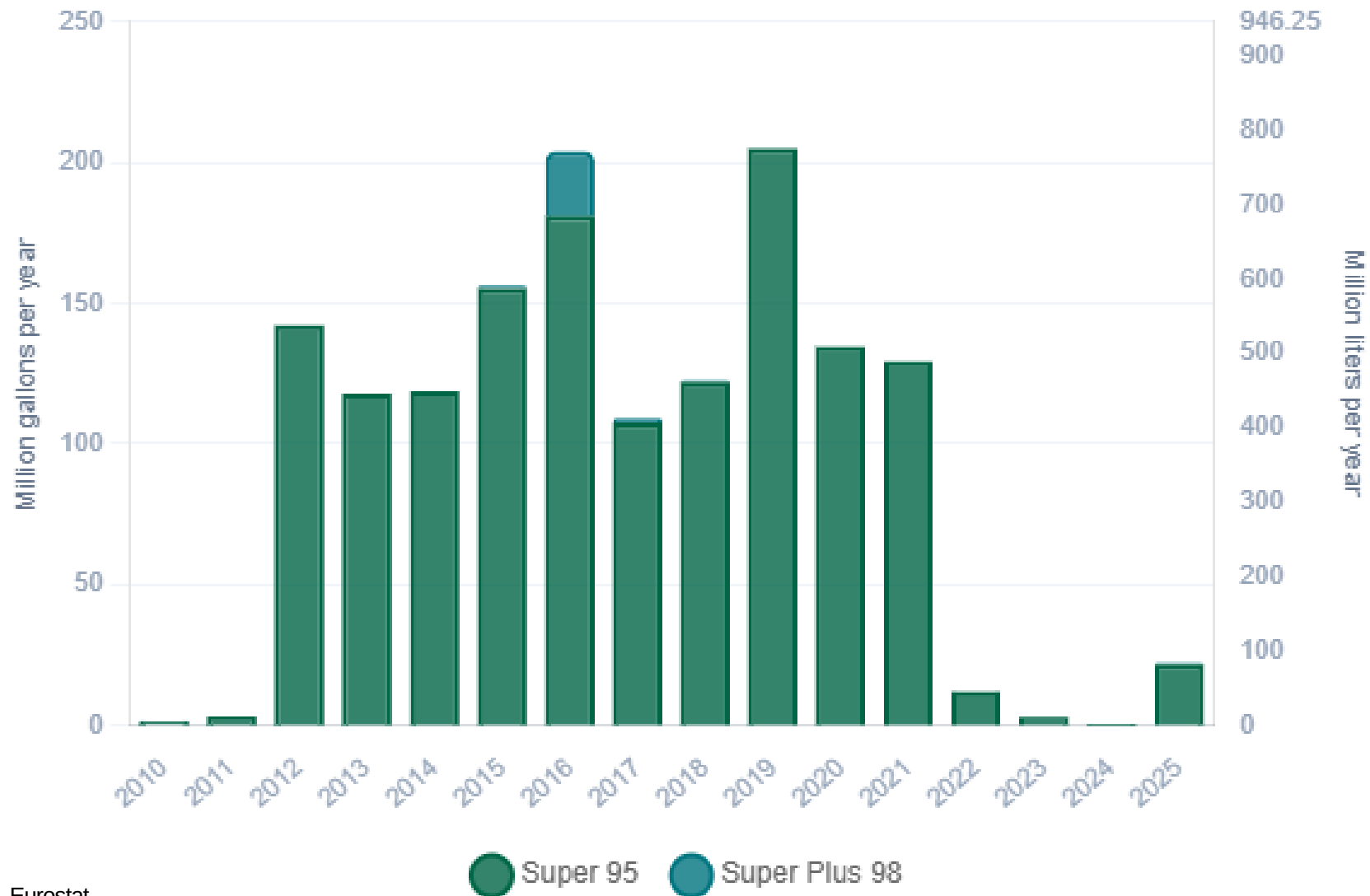
Source: Eurostat

Gasoline Production in Greece



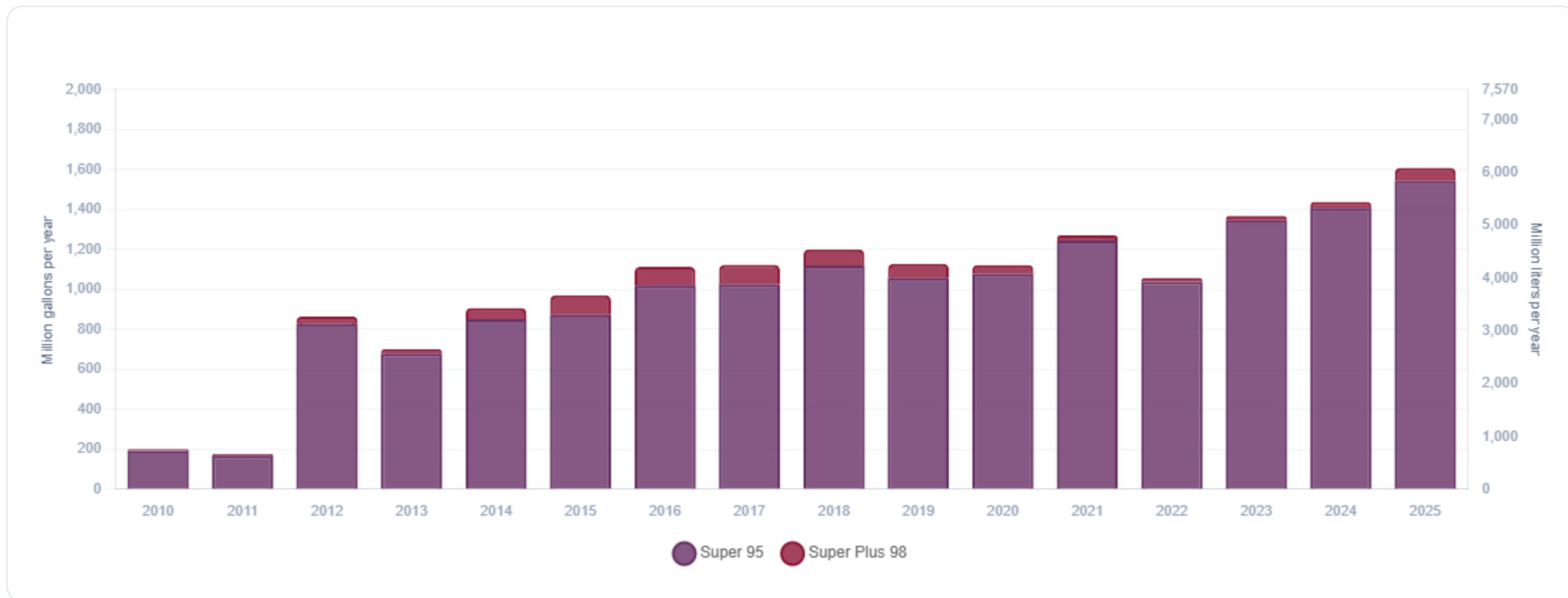
Source: Eurostat

Gasoline Imports in Greece



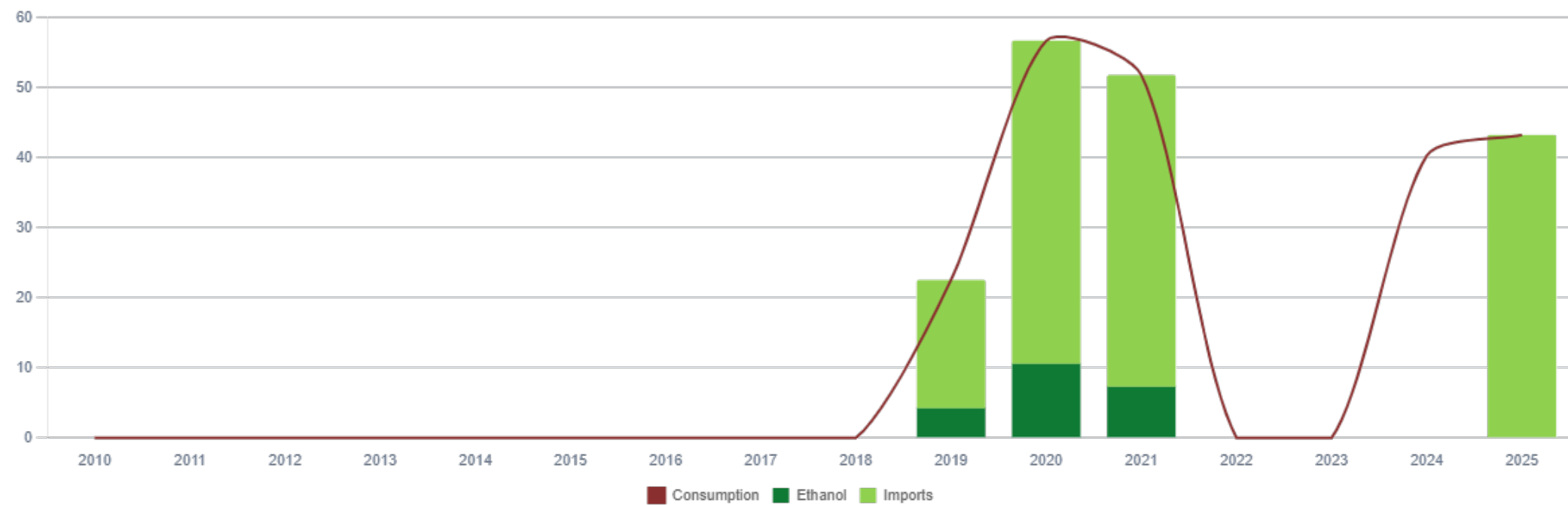
Source: Eurostat

Gasoline Exports in Greece



Source: Eurostat

Ethanol Balance in Greece



Source: Eurostat, USDA – Fuel Mandates in EU

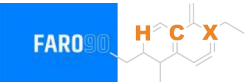
Gasoline Quality in Greece

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	3.3							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

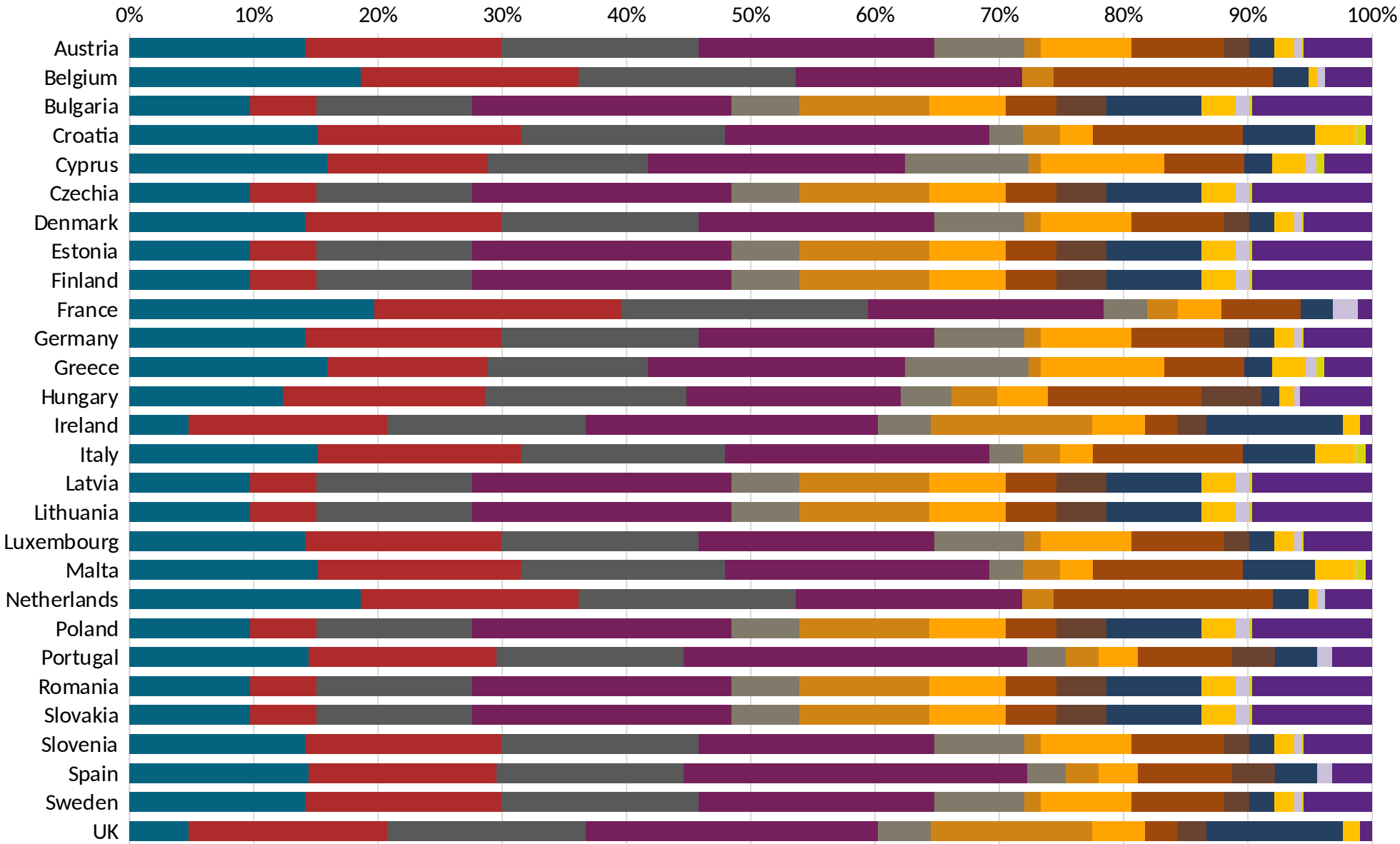
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



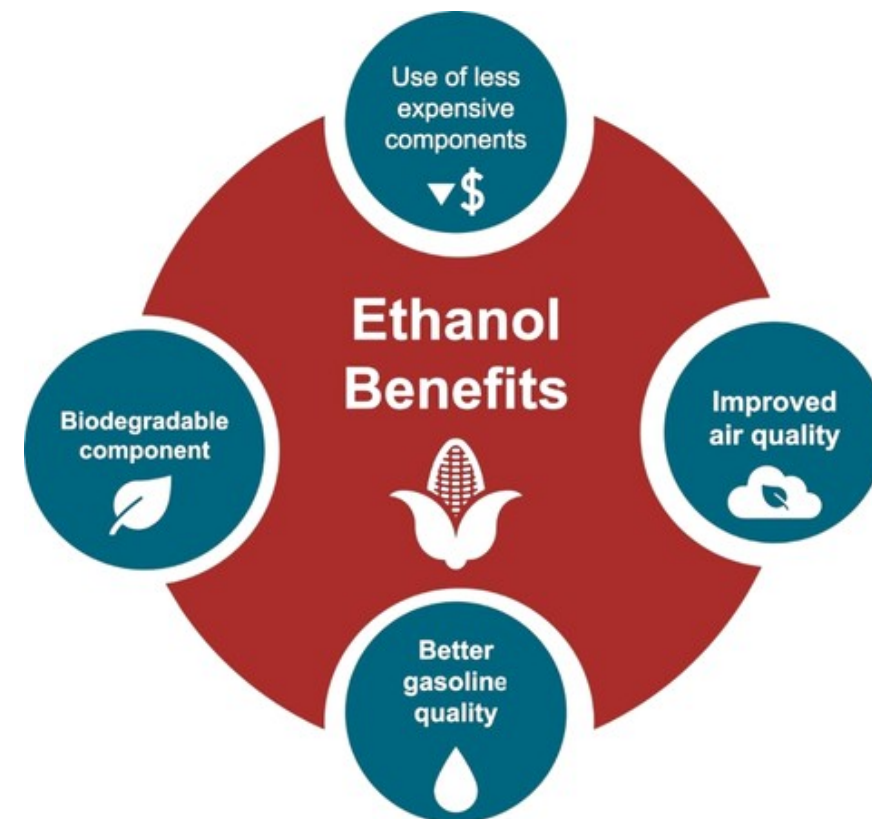
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

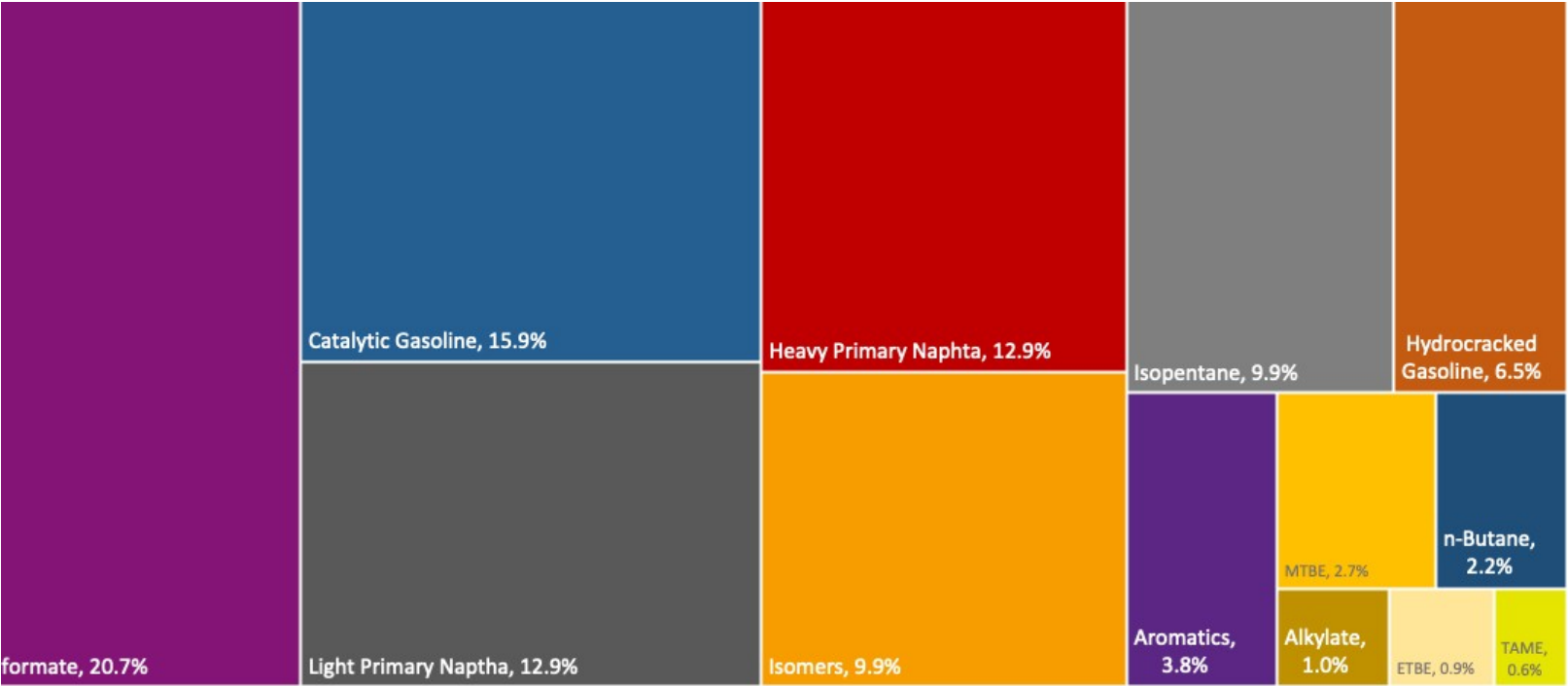
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

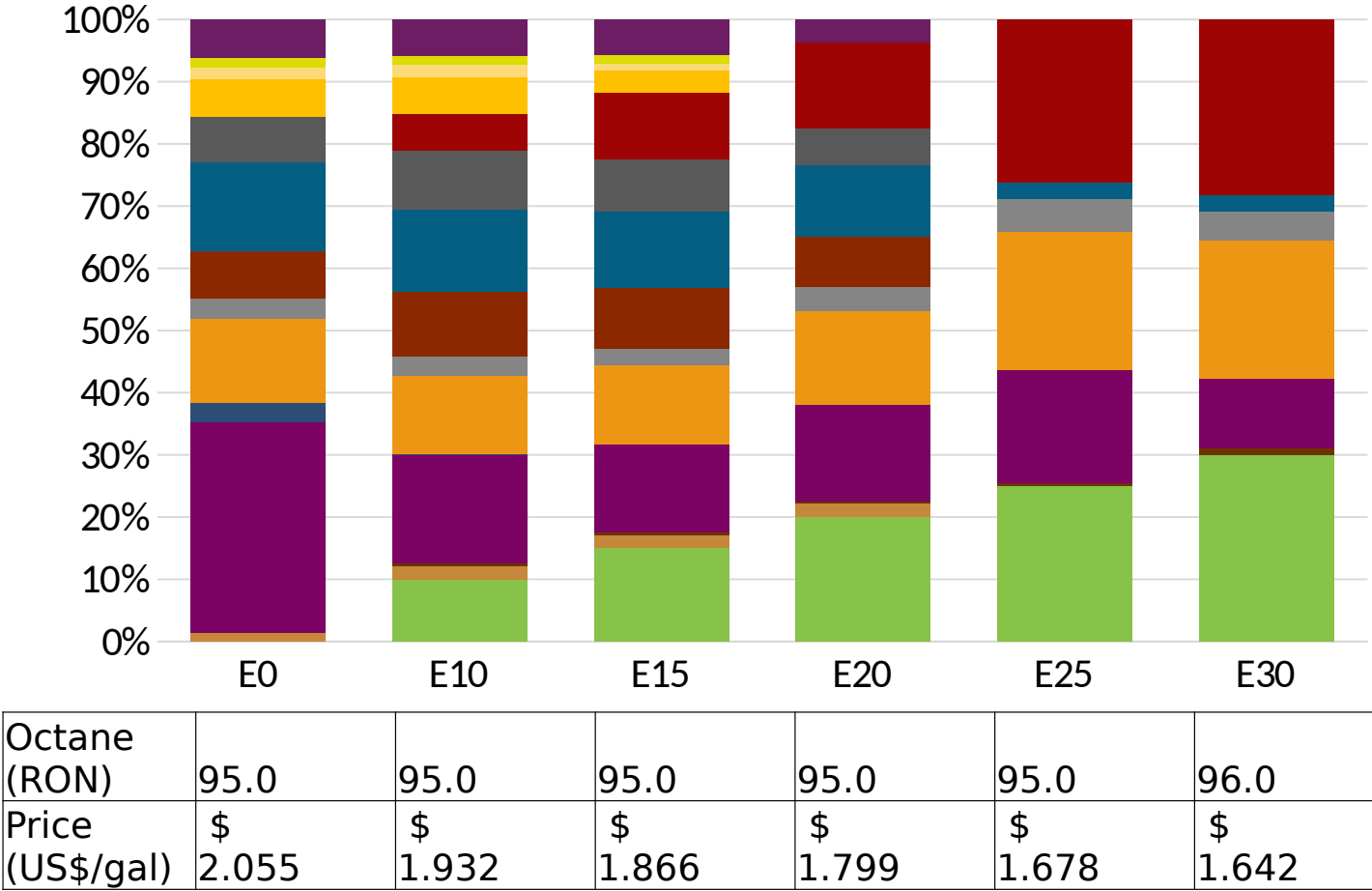
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Greece Available Blending Components

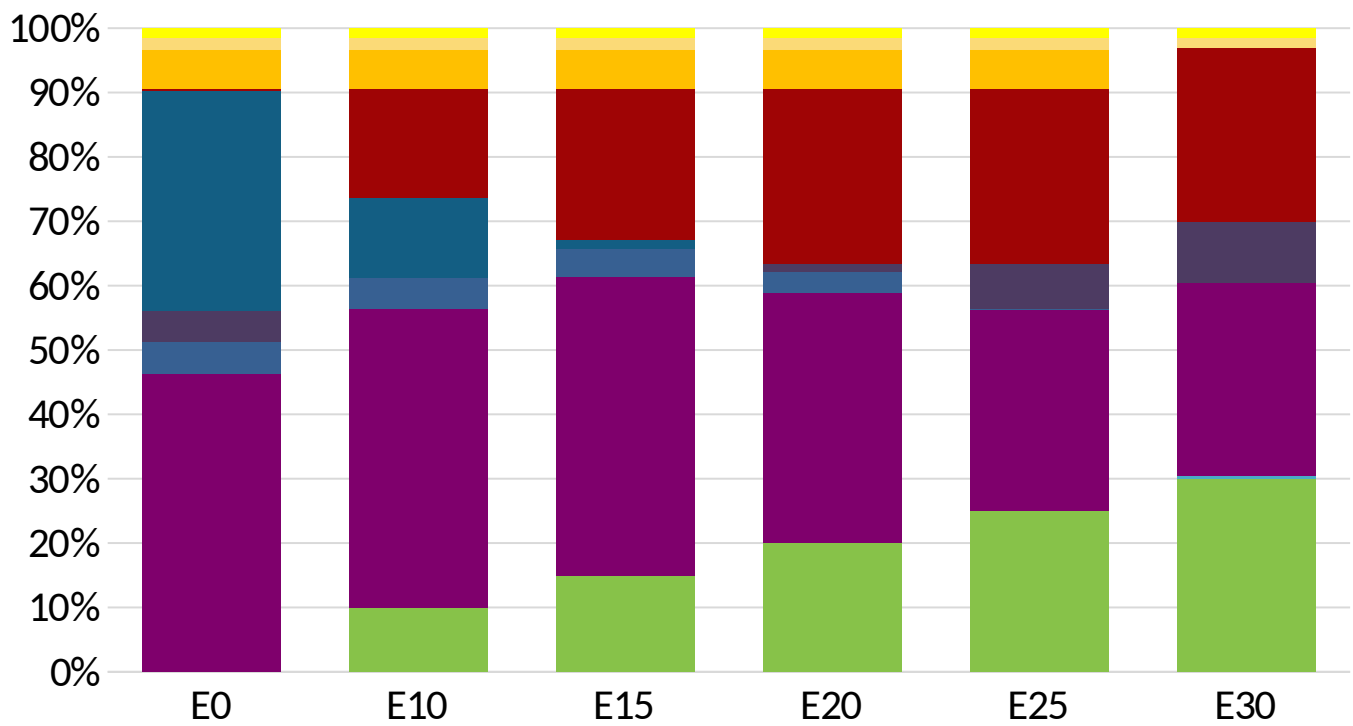
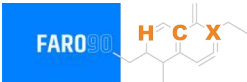


Greece – Gasoline 95 – Constant Octane



Average CIF 2025 Prices

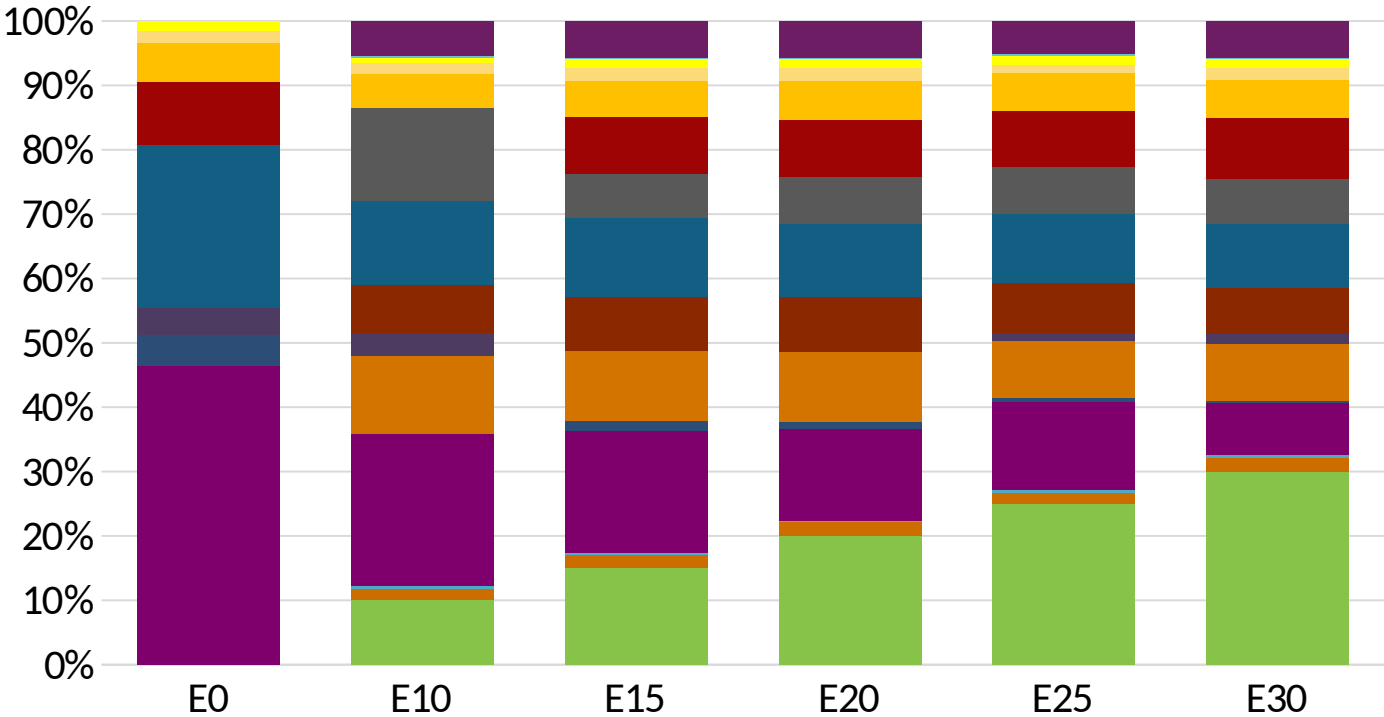
Greece - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.127	\$ 2.001	\$ 1.929	\$ 1.854	\$ 1.777	\$ 1.721

Average CIF 2025 Prices

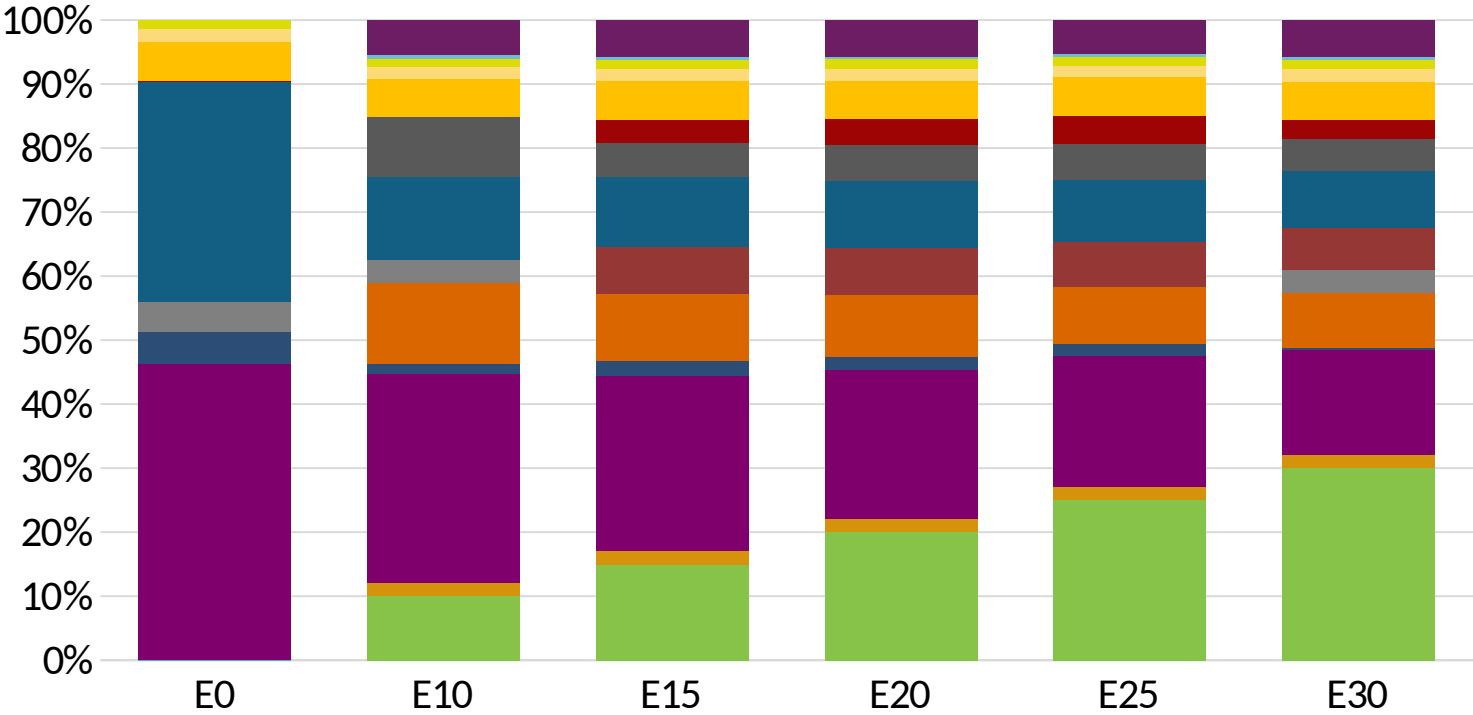
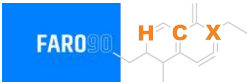
Greece – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	95.0	96.2	97.9	99.4	100.8	102.2
Price (US\$/gal)	\$ 1.993	\$ 1.993	\$ 1.993	\$ 1.993	\$ 1.993	\$ 1.993

Greece – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	99.4	100.7	102.0	103.3	104.5
Price (US\$/gal)	\$ 2.127	\$ 2.127	\$ 2.127	\$ 2.127	\$ 2.127	\$ 2.127

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

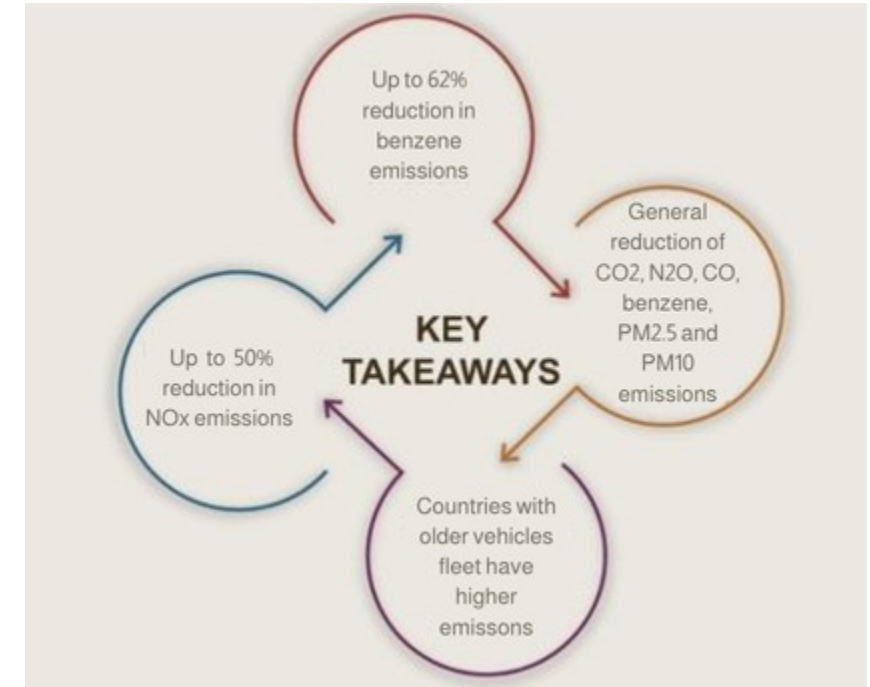
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

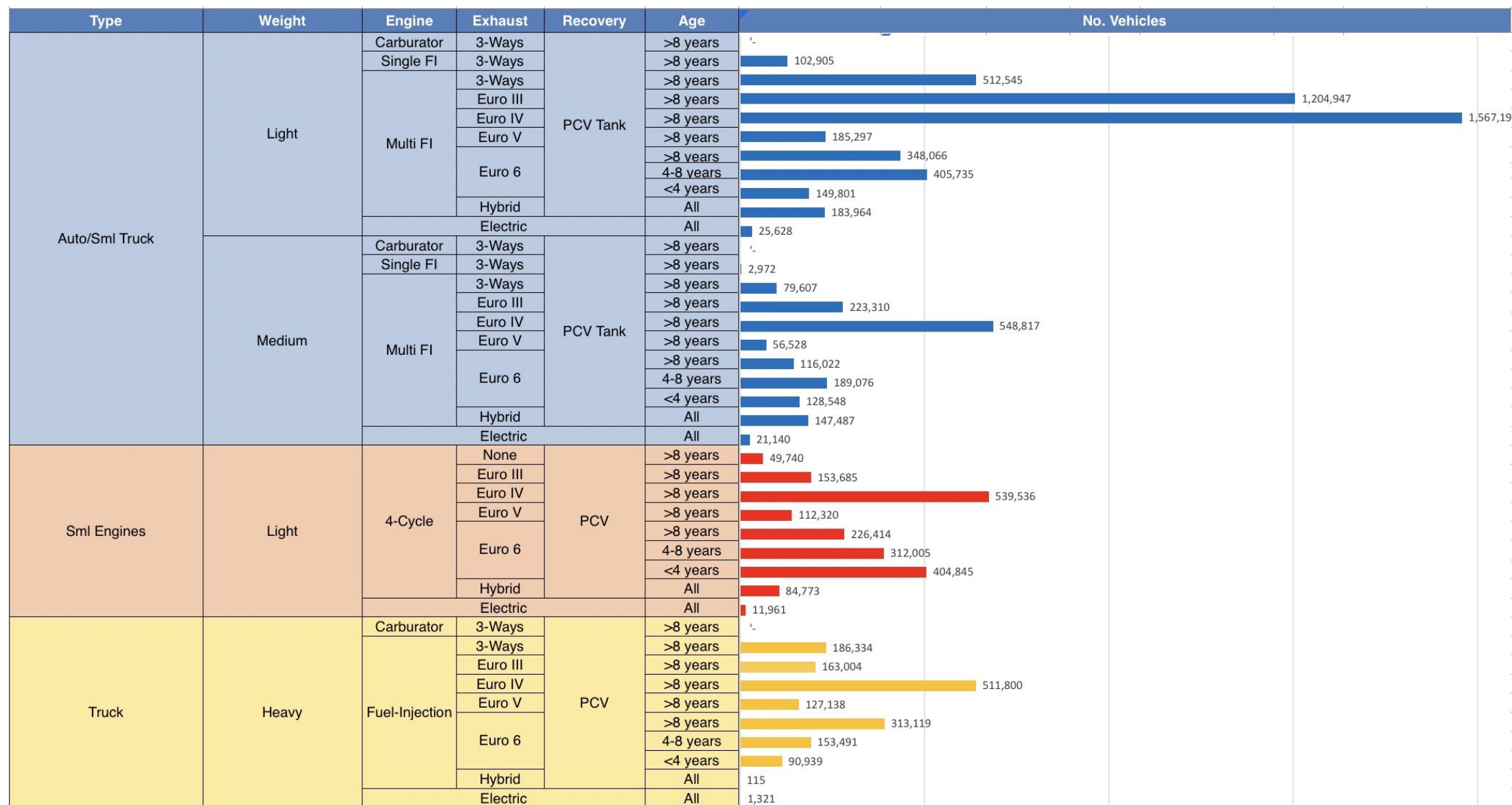
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Greece

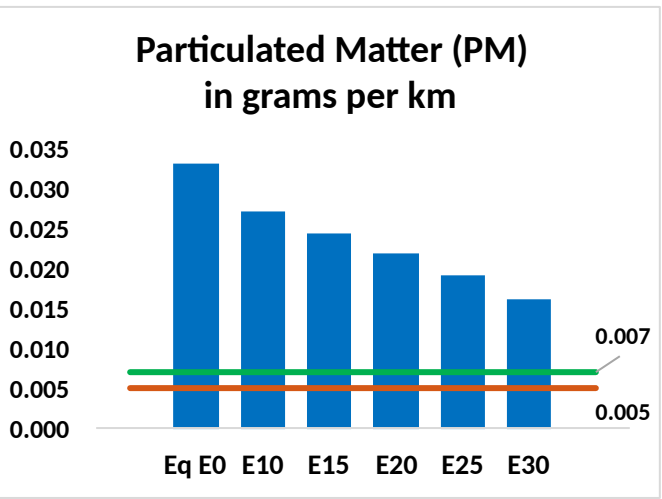
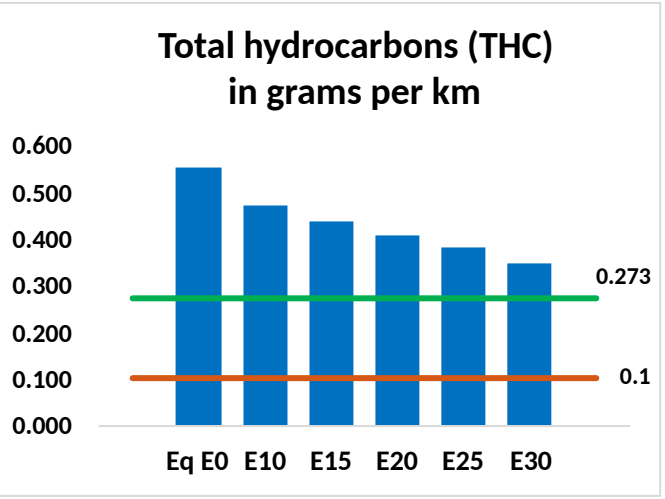
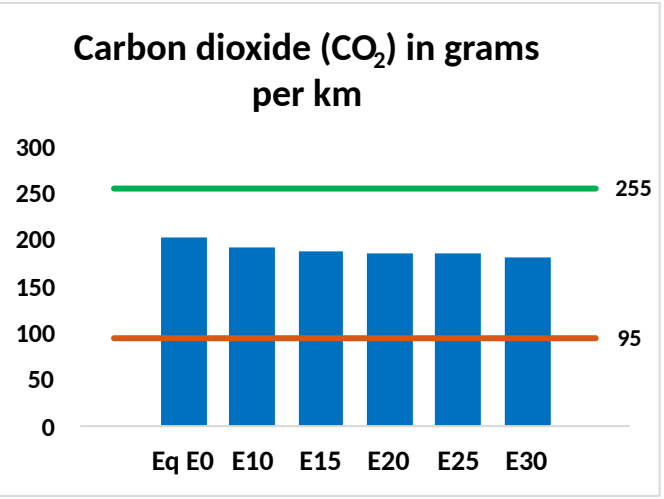
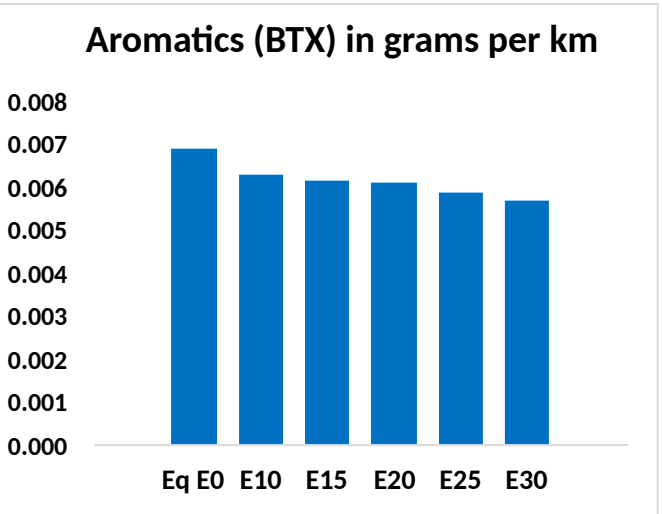
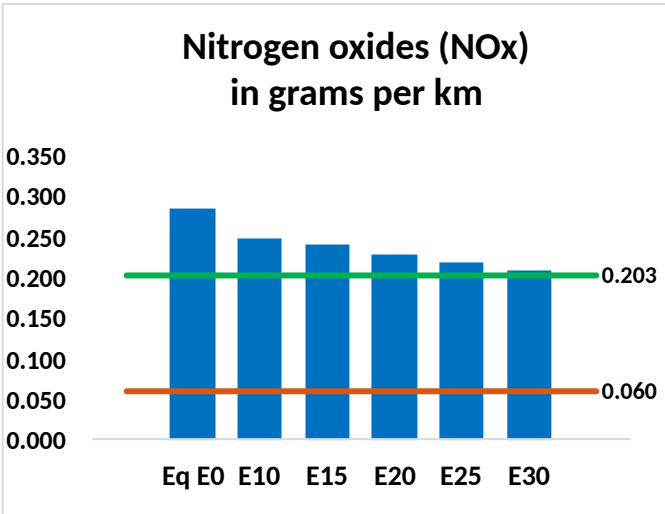
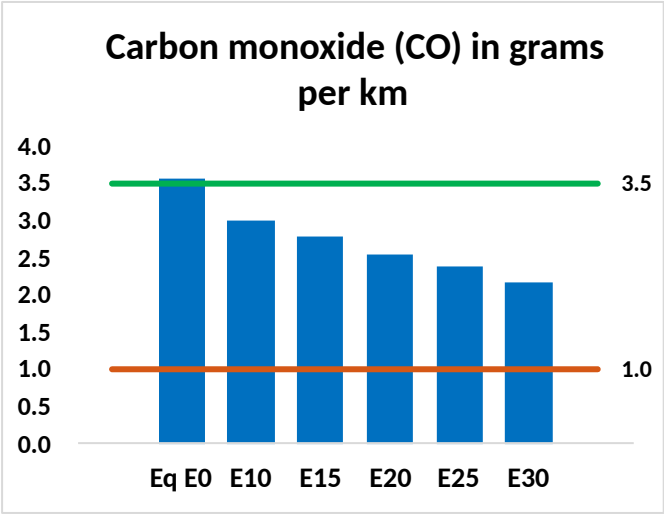
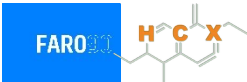


Vehicle Fleet: 9,642,126
Gasoline Vehicle Fleet: 6,833,958

Source: Eurostat, ACEA

Greece – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Greece – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	566,219	522,014	477,809	441,453	404,806	376,609	345,253	-16%	-29%
CO2	32,196,010	31,368,113	30,586,209	29,971,266	29,666,967	29,499,709	28,955,945	-5%	-8%
VOC	88,038	81,630	75,221	70,040	64,858	60,883	55,544	-15%	-26%
NOx	45,556	42,063	39,681	38,476	36,525	35,000	33,202	-13%	-20%
SOx	365	338	310	287	264	240	214	-15%	-28%
NH3	11,790	11,674	11,557	11,612	11,742	11,834	11,911	-2%	0%
N2O	442	440	438	440	449	454	459	-1%	2%
CH4	19,149	19,149	19,149	19,436	19,857	20,221	20,566	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	579	556	533	518	505	496	481	-8%	-13%
Acetaldehyde	1,629	1,870	2,743	4,178	5,692	6,504	7,685	68%	249%
Formaldehyde	5,855	5,692	6,636	7,792	8,163	9,030	9,831	13%	39%
Benzene	1,099	1,056	1,001	985	977	934	904	-9%	-11%
PM2.5	1,932	1,756	1,581	1,426	1,275	1,113	943	-18%	-34%
PM10	3,366	3,060	2,754	2,484	2,221	1,939	1,643	-18%	-34%

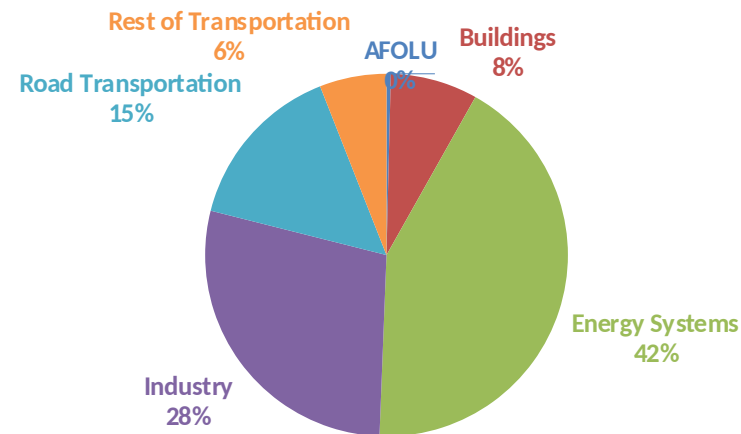
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

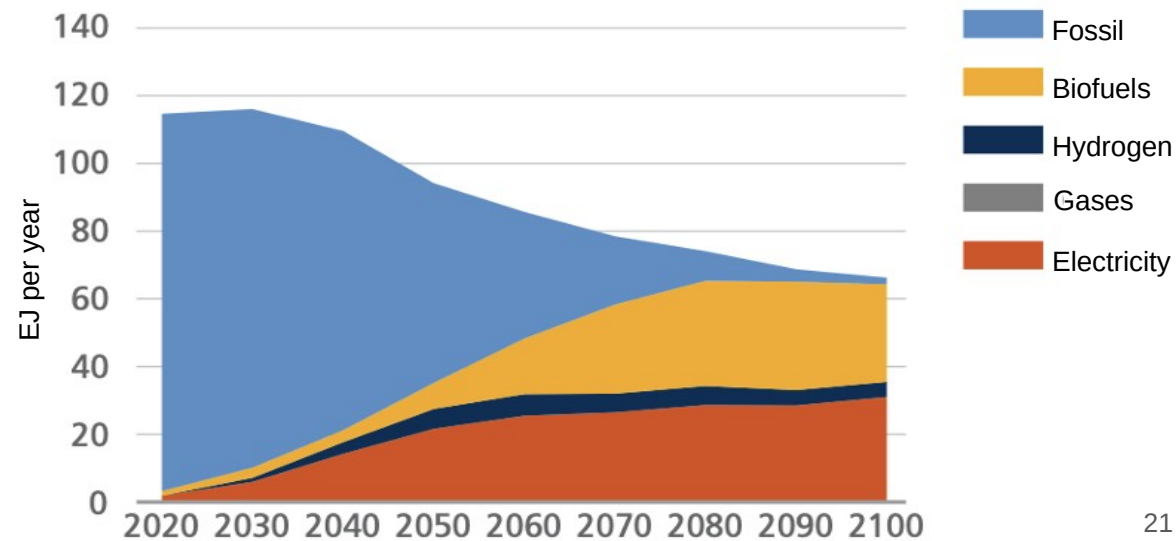
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

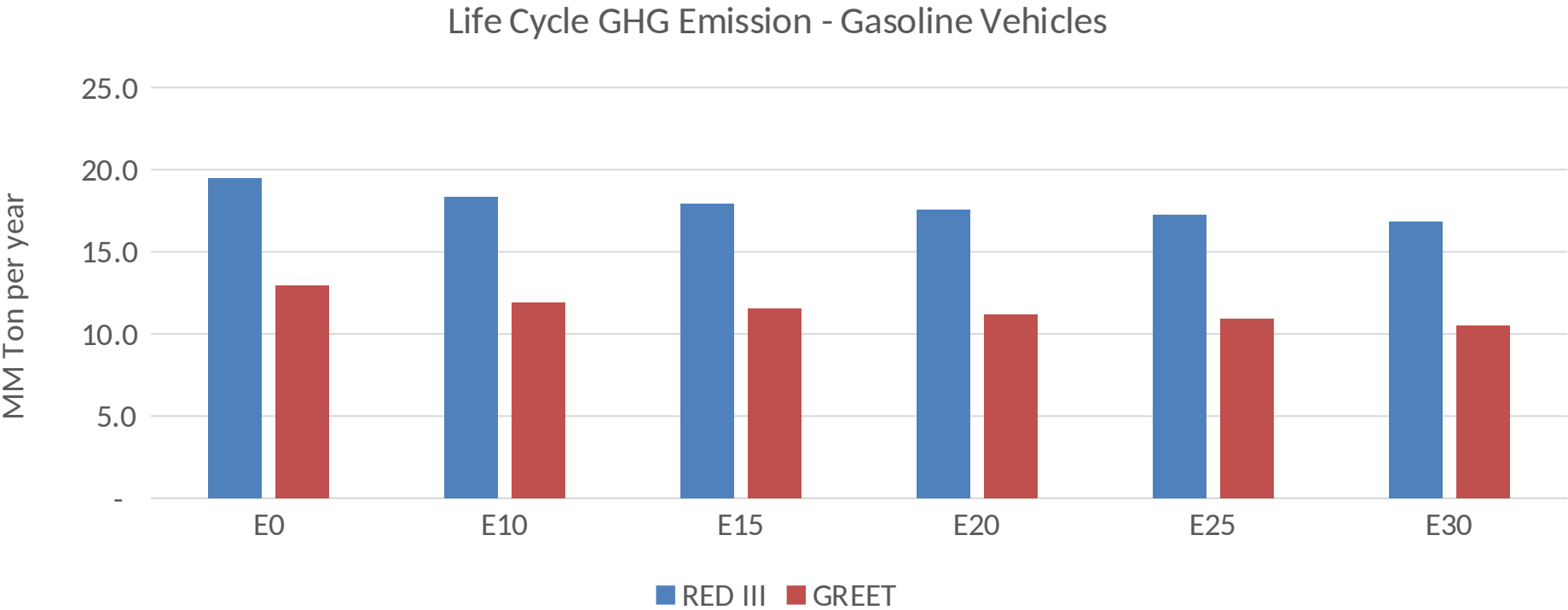
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Greece – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	74.3
Target @ 2035 (MMT/year)	41.8
Delta	32.5